

Towards Explainable AI in Nuclear: Introducing Ad Hoc Model Explainability

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What is Explainable AI?

- *Explainability* refers to the degree to which we can understand *why* a model arrived at its conclusion.
  - E.g.: *Why* did the model tell us Olive is a dog?
    - Olive has long ears
    - Olive has a wide, brown nose
    - Olive has a long snout
- *Interpretability* refers to the degree to which we can understand *how* a model arrived at its conclusion.
  - E.g.: *How* did the model determine that Olive is a dog?
    - **IF** *ears > 3 in.* **AND** *nose > 1 in*. **AND** *snout > 3 in.* **THEN** dog
- We need both model explainability and interpretability to gain model trust.
- We can gain explainability in post; interpretability requires interpretable models themselves.

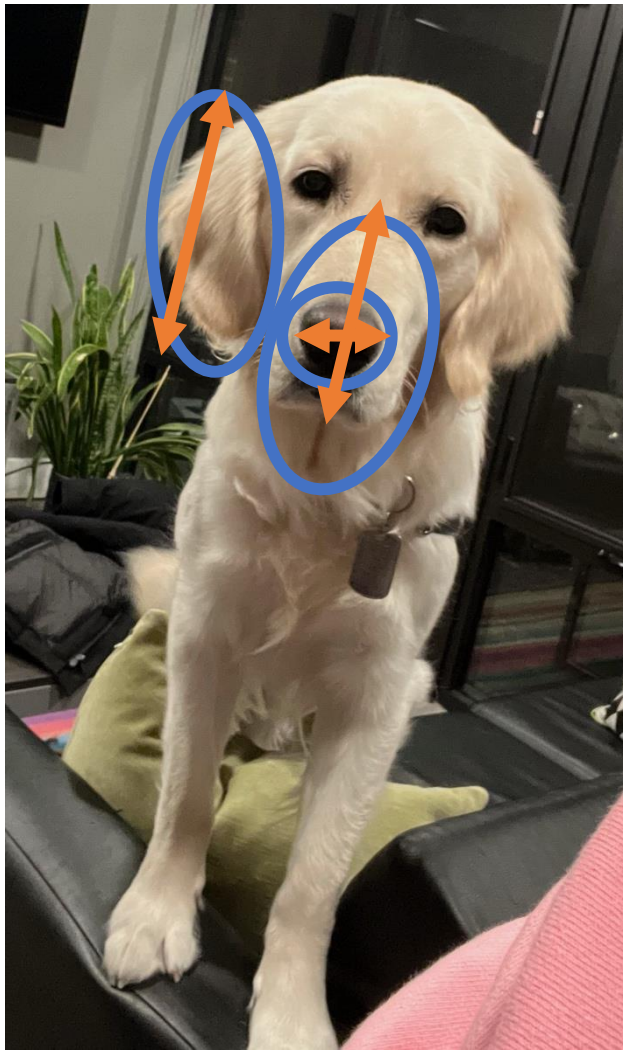


Fig. 1: Olive, a service dog in training at the University of Illinois Urbana-Champaign.

Motivation

- We know there are lots of benefits to using AI, but nuclear (and other sensitive industries) often **cannot afford to be wrong**
- Explainable (and interpretable) AI methods help increase our trust in a machine-learning model
- The U.S. Nuclear Regulatory Commission outlines this need in their Artificial Intelligence Strategic Plan for fiscal years 2023-2027 (NUREG-2261)
- Other domestic and international federal agencies are requesting this as well (U.S.: Federal Aviation Administration, Germany, China)
- A case study: critical heat flux (CHF)
  - No known closed-form solution
  - Limits the power level of a reactor
  - But an underestimate could be catastrophic

Can we predict CHF using a feedforward neural network and then explain our results using ad hoc model explainability to evaluate the model?

SHAP Theory

- A unified framework that provides feature attributions for model predictions
- It is based on cooperative game theory, which allocates the contribution of each player (i.e., model feature) fairly based on their participation in a “game” (i.e., model prediction).
- It considers all possible subsets of features and calculates the contribution of each feature towards the final prediction (+/-)
- We consider three flavors of SHAP:
  - Classic
    - Exact method, as shown by Equation 1. Increases in complexity as $O(d \cdot 2^d)$ where d is the number of features.
  - Deep Learning Important Features (DeepLIFT)
    - Uses ”reference differences” to assess the relative importance of each input feature instead of traditional gradients
    - Uses multipliers to propagate contributions through each neuron in the network
  - Integrated Gradients
    - Integrates gradients of the model’s output concerning a specific input along a path from a stated baseline (often zero) to the actual input.
- *See Lundberg et al. [11] for more details on SHAP theory.*

Classic SHAP

Equation 1

$$\phi_i = \sum_{S \subseteq \{1,2,\ldots,d\} \setminus \{i\}} \frac{|S|!(d - |S| - 1)!}{d!} [f(S \cup \{i\}) - f(S)]$$

where

$$d = \text{the number of features in the model}$$

$$\frac{|S|!(d - |S| - 1)!}{d!} = \text{the weight for each subset S}$$

$$f(S) = \text{the model's prediction using only the features in the subset S}$$

$$f(S \cup \{i\}) - f(S) = \text{represents the marginal contribution of player i to the team S}$$

DeepLIFT

Equation 2

$$\text{DeepLIFT}_i(x) = (x_i - x_i') \cdot \text{Multiplier}_i$$

where:

$$\text{Multiplier}_i = \frac{\Delta f}{\Delta x_i} = \frac{f(x) - f(x')}{x_i - x_i'}$$

and:

$$\begin{aligned}
x_i &= \text{the actual input value of feature i,} \\
x_i' &= \text{the reference (baseline) input value of feature i,} \\
f(x) &= \text{the output of the neural network for the actual input x,} \\
f(x') &= \text{the output of the neural network for the baseline input x',} \\
\Delta f &= \text{the change in the network output,} \\
\Delta x_i &= \text{the change in the input feature i,} \\
\text{Multiplier}_i &= \text{a term representing the contribution of the change in input feature i} \\
&\quad \text{to the change in the network output.}
\end{aligned}$$

Integrated Gradients

Equation 3

$$\mathrm{IG}_i(x) = (x_i - x_i') \int_{\alpha=0}^{1} \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} \, d\alpha$$

where:

x = the actual input vector,

x' = the baseline input vector, often chosen as a zero vector or some neutral input,

x_i = i-th feature of the actual input,

x_i' = i-th feature of the baseline input,

f = the function representing the model output,

α = a scaling parameter ranging from 0 to 1,used to interpolate between the baseline and the actual input,

$\frac{\partial f(x)}{\partial x_i}$ = the gradient of the model output fwith respect to the input feature x_i.

Methodology

- Dataset: 60 years of experimental CHF data from an OECD Nuclear Energy Agency + U.S. Nuclear Regulatory Agency collaboration
  - 21,435 data points
  - Features: test section diameter (D), heated length (L), pressure (P), mass flux (G), inlet temperature (Tin), outlet quality (Xe)
  - Output: Critical heat flux (CHF)
- Generated a feedforward neural network (FNN) using hyperparameters tuned using Bayesian optimization via pyMAISE [10]
- Compared to a *more* interpretable model, a random forest

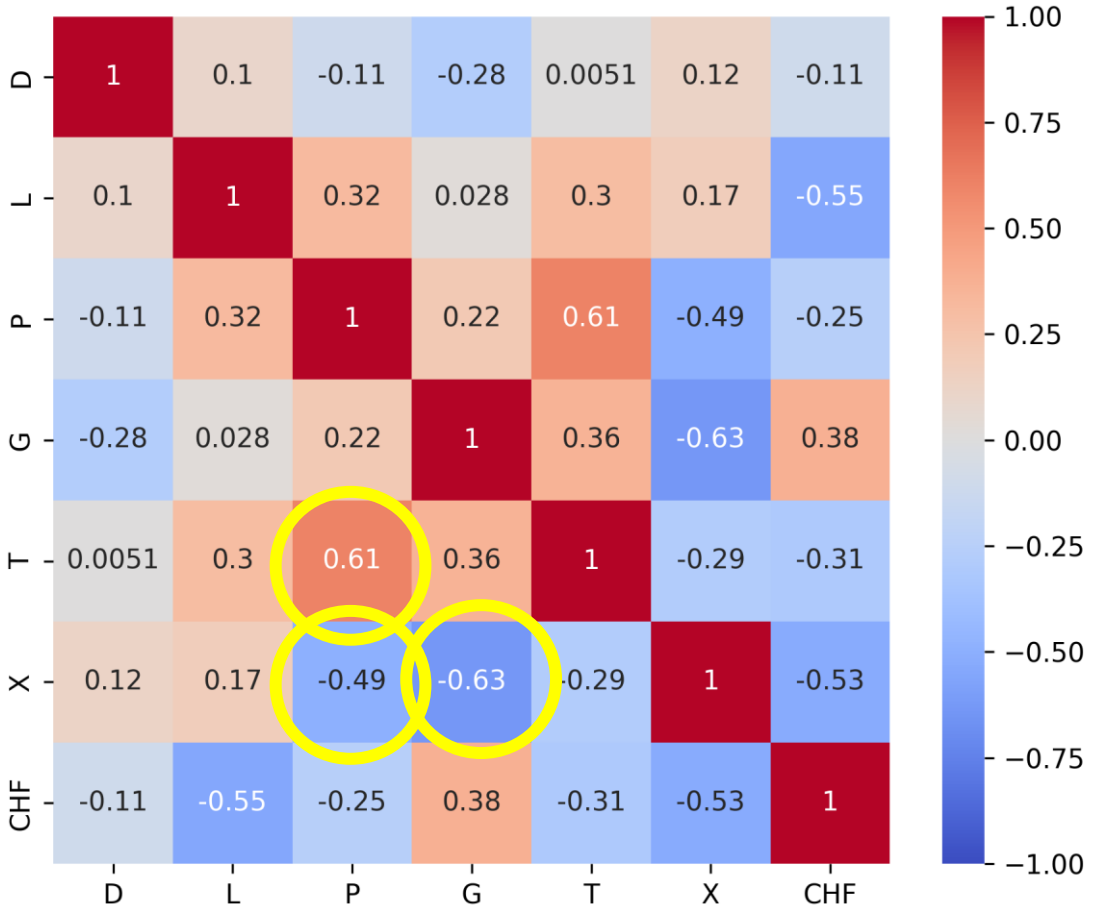


Fig. 2: Correlation matrix between CHF and the full set of input features available in the dataset

Random Forest Validation

- As a validation case, we applied a **permutation importance (PI)** algorithm and a **mean decrease in impurity (MDI)** algorithm on a random forest model (R^2 =0.993) that performed nearly as well as our FNN (R^2 = 0.997) [10].
- Permutation Importance
  - The values of each feature in the dataset are randomly permuted (one column at a time) and the random forest is evaluated on the shuffled dataset– repeated to reduce variance
  - High decrease in accuracy indicates high feature importance
- Mean Decrease in Impurity
  - Measures the importance of each feature by examining impurity it causes across all the decision trees in the forest
  - When a decision tree splits on a feature (x_i), the impurity of the resulting child nodes is generally lower than that of the parent node.
  - The difference in impurity before and after the split is known as the decrease in impurity.
  - The higher the MDI, the higher the importance of the feature.

Isn't feature importance *good enough?*

- Feature importance doesn't give contributions for individual samples
- Some feature importance algorithms (mean decrease in impurity) can overstate the importance of numerical features
  - This is true even if the numbers are meaningless, like a serial number
- Feature importance methods can be easily skewed by overfitting
- SHAP gives us much more information and, therefore, insight, about how the model works

Exact vs. DeepLIFT

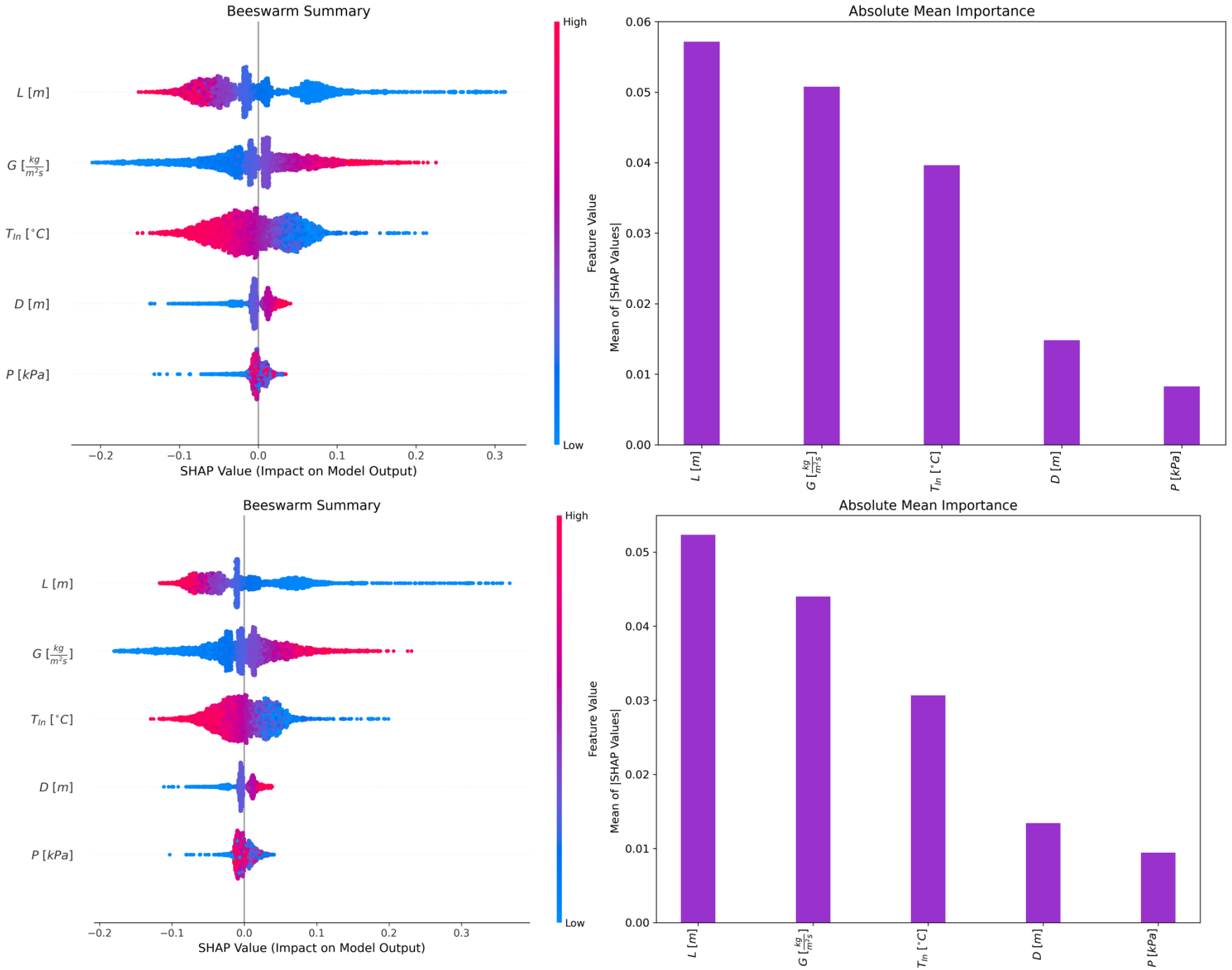


Fig. 3: Explainability results using Exact SHAP (top) and DeepLIFT (bottom) for the best CHF FN model with all features *except* outlet quality.

Correlated Features

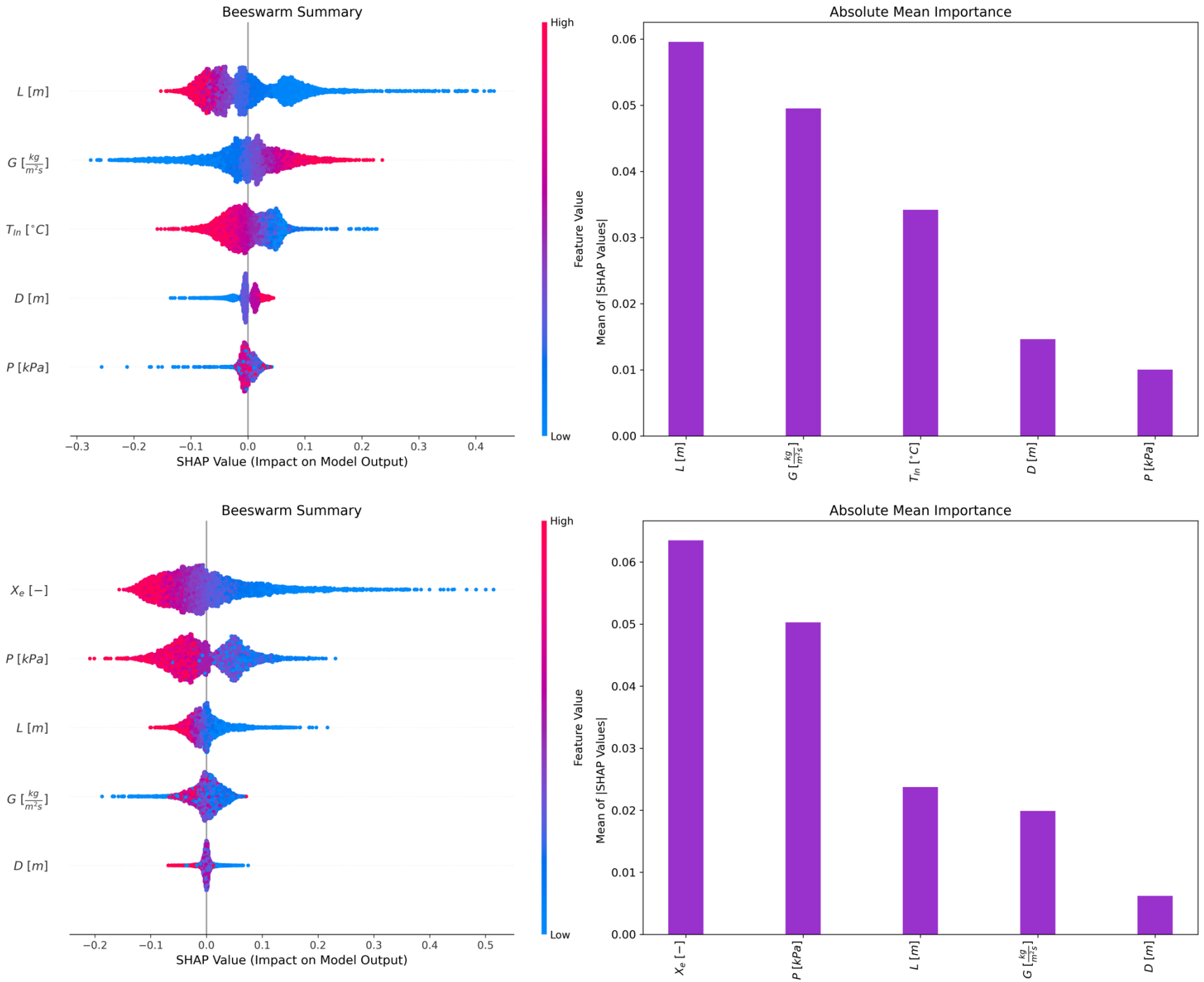
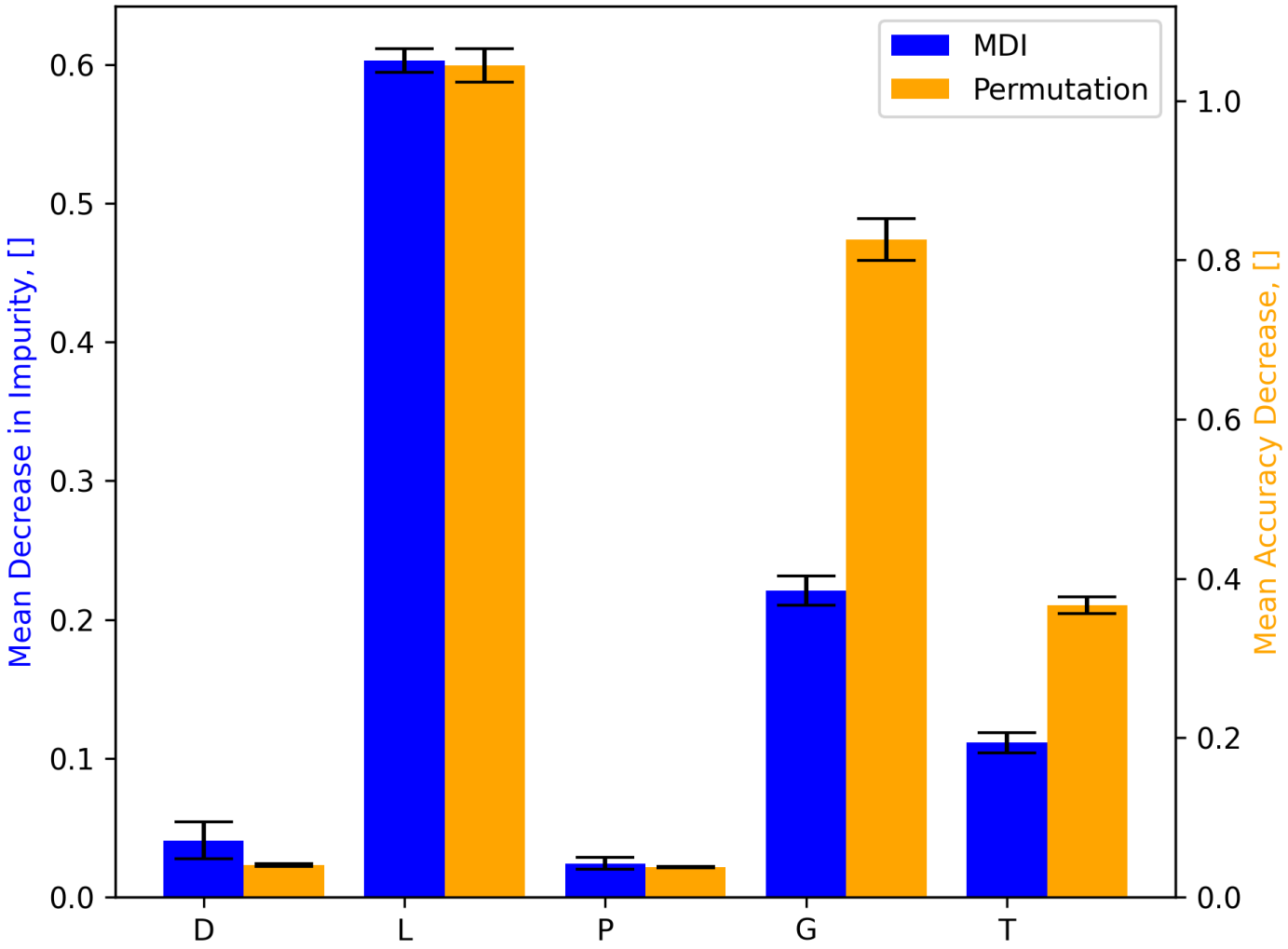


Fig. 4: Explainability results using Integrated Gradients for the best CHF FNN model with all features *except* outlet quality (Xe) (top) and inlet temperature (Tin) (bottom).

Random Forest Importances



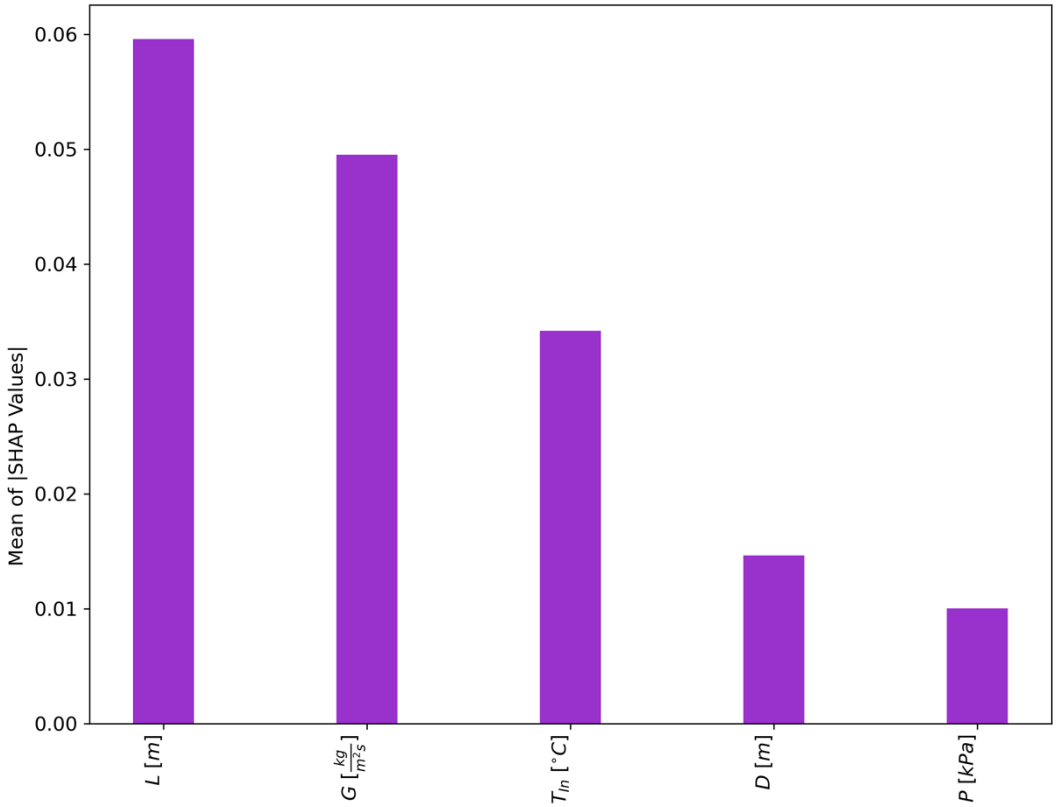


Fig. 5: Feature importances using permutation importances and mean decrease in impurity for a random forest model with all features except outlet quality (left). Repeat of IG results for FNN (top).

Additional Datasets – Boiling Water Reactor

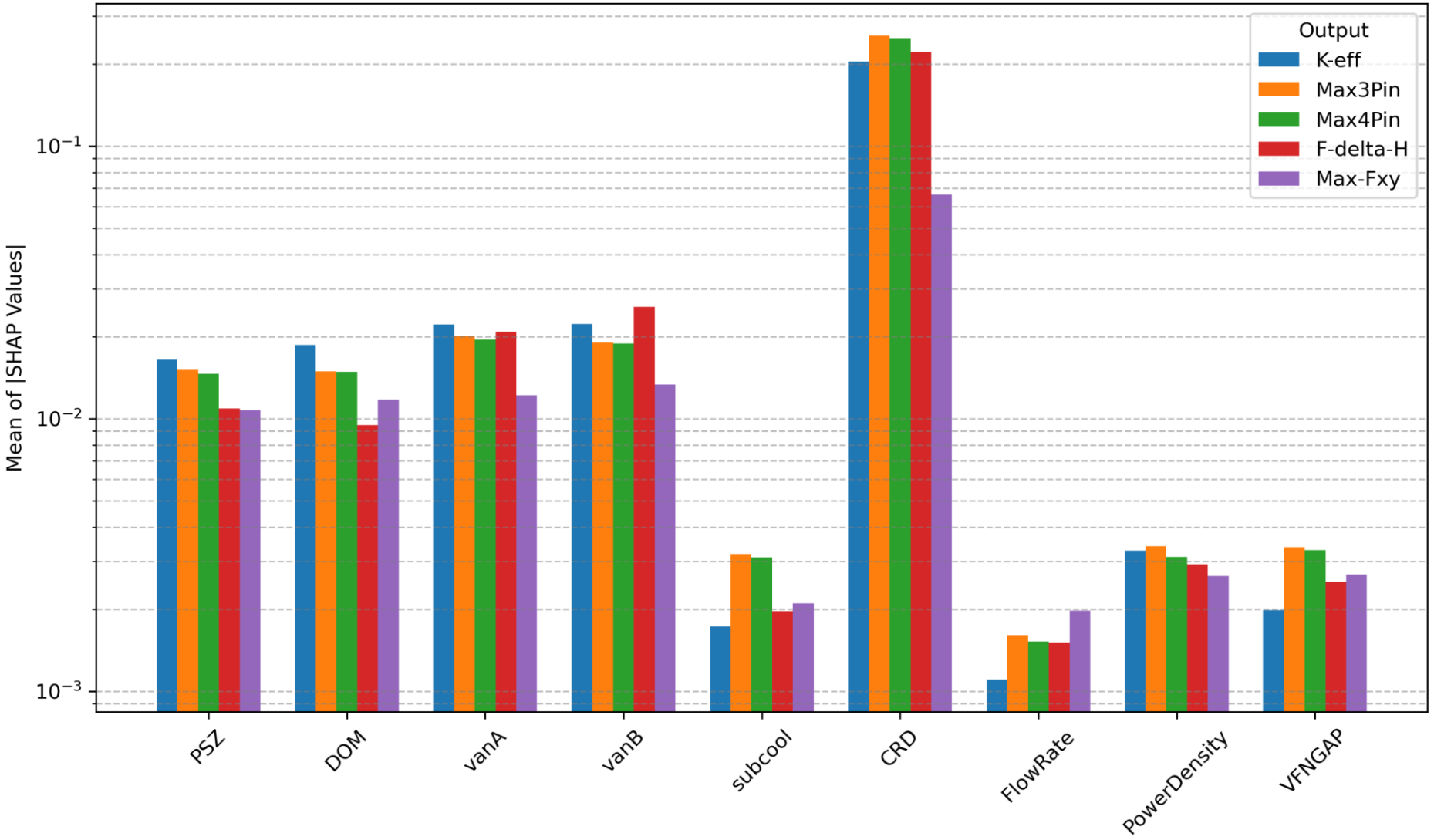


Fig. 6: Mean of absolute SHAP values for boiling water reactor simulated dataset using Kernel SHAP (weighted kernel + local linear approximation LIME) on FNN generated by Myers et al. [10].

Additional Datasets – Fuel Performance

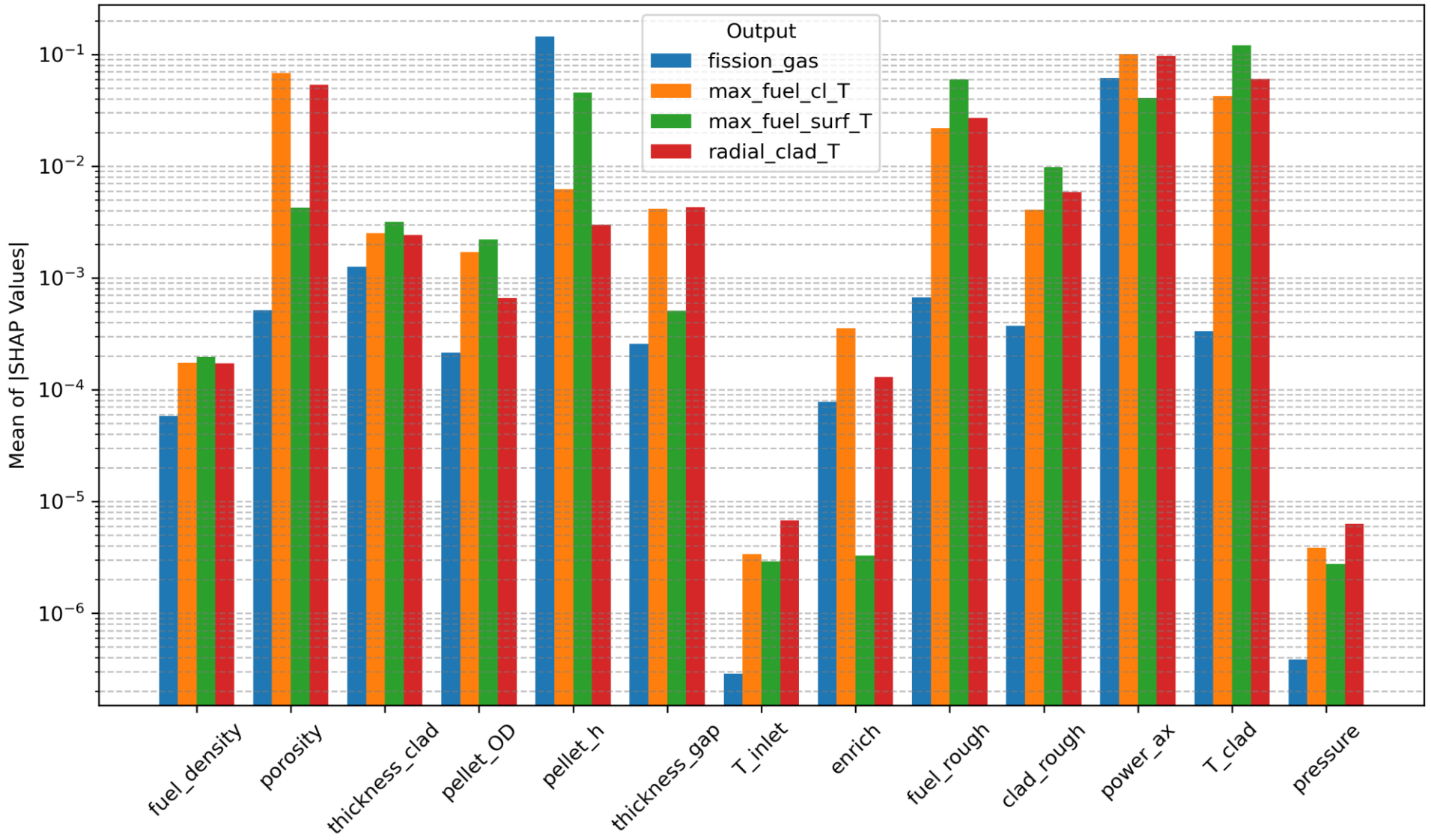


Fig. 7: Mean of absolute SHAP values for fuel performance simulated dataset using Kernel SHAP (weighted kernel + local linear approximation LIME) on FNN generated by Myers et al. [10] using data originally presented by Radaideh and Kozlowski [15].

Conclusions & Future Work

- DeepLIFT, Integrated Gradients (IG), Classic SHAP, and random forests had consistent feature ranking in predicting CHF
- Mass flux and channel length emerged as the most important features for predicting CHF
  - This aligns with what we expect based on existing correlations
- Future work is needed to improve model interpretability and layer/node attribution-based explainability
- Models with higher inherent interpretability yield more trustworthy explainability results, future work should search for high performing, interpretable models

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Questions?